Huawei Access Node Hardware/Environmental Fault Troubleshooting SOP (BAS-ACC-07)

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2049

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve a hardware/environmental fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) on a Huawei Access Node (BAS-ACC-07) located at the Basavanagudi POP. The failure is detected by an ALARM_2049 alarm with a severity of Minor. The root cause of this failure domain is primarily attributed to the inherent hardware design of the Huawei Access Node, which relies on the redundancy of critical components such as power supplies and line cards. In the event of a power supply redundancy loss or line card crash, the node will trigger the ALARM_2049 alarm. This alarm is generated by the node's fault management system, which detects the failure and notifies the network operator. The underlying protocol mechanics that contribute to this failure domain include the Huawei Access Node's use of ASICs (Application-Specific Integrated Circuits) for packet processing and the reliance on line cards for network connectivity. The environmental factors that may contribute to this failure domain include power supply voltage fluctuations, high temperatures, and humidity.

Pre-requisite Checks

- Check: Verify the site location and node ID to ensure the correct node is being troubleshooted (BAS-ACC-07 at Basavanagudi POP).

- Check: Confirm the ALARM_2049 alarm is present and has a severity of Minor.

- Check: Verify the power supply redundancy status and line card connectivity to ensure the node is functioning correctly.

Potential Root Causes

Cause	Probability
Power supply redundancy loss	High
Line card crash	Medium
ASIC memory parity errors	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-management alarm, Expected Output: ALARM_2049: Minor, Power supply redundancy loss detected, Description: This command is used to verify the presence of the ALARM_2049 alarm and determine the alarm's severity.

Command: display power supply redundancy, Expected Output: Power supply redundancy: Lost, Description: This command is used to verify
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the power supply redundancy status and determine if the node is
functioning correctly.

Command: display line card, Expected Output: Line card 0/0/1: Crash
detected, Description: This command is used to verify line card
connectivity and determine if the node is functioning correctly.
```

Resolution Steps

Step 1: Step: Verify power supply redundancy, Command: display power supply redundancy, Description: Verify the power supply redundancy status and replace the faulty power supply if necessary.

Step 2: Step: Verify line card connectivity, Command: display line card, Description: Verify line card connectivity and replace the faulty line card if necessary.

Step 3: Step: Reset the node, Command: reset system, Description: Reset the node to clear any faults and alarms.

Post-Check Verification

```
Command: display fault-management alarm, Expected Output: No alarms
present, Description: Verify that no alarms are present after
resolution.

Command: display power supply redundancy, Expected Output: Power
supply redundancy: Normal, Description: Verify the power supply
redundancy status after resolution.

Command: display line card, Expected Output: Line card 0/0/1: Normal,
Description: Verify line card connectivity after resolution.
```

Roll-back Procedures

- Step: Revert power supply redundancy changes, Command: undo power supply redundancy, Description: Revert any changes made to the power supply redundancy status.

- Step: Revert line card changes, Command: undo line card, Description: Revert any changes made to the line card connectivity.